

A Time-Series Analysis: The Duration and Impact of Productivity Gains From New Technology

Reading Notes - The Duration and Impact of Productivity Gains From New Technology

▼ Sources:

- [1] <https://www.forbes.com/sites/janakirammsv/2026/05/17/uber-burns-its-2026-ai-budget-in-four-months-on-claude-code/>
- [2] <https://byteiota.com/uber-blows-2026-ai-budget-on-claude-code-in-4-months/>
- [3] <https://www.businessinsider.com/ai-spending-roi-concerns-tokenmaxxing-uber-coo-andrew-macdonald-reaction-2026-5>
- [4] <https://jacobin.com/2026/06/ai-anthropic-openai-costs-financing>
- [5] <https://www.library.hbs.edu/working-knowledge/many-gen-ai-players-remain-far-away-from-profiting-interview-with-andy-wu>
- [6] <https://garymarcus.substack.com/p/slop-productivity-and-why-the-ai>
- [7] https://www.marketwatch.com/story/big-techs-700-billion-spending-on-ai-this-year-is-called-the-greatest-capital-misallocation-in-history-7d44aa4b?mod=dist_amp_social&link=sfmw_tw
- [8] <https://www.cnn.com/2026/04/17/ai-tokens-anthropic-openai-nvidia.html>
- [9] <https://freedium-mirror.cfd/https://medium.com/business-as-usual-io/tokenmaxxing-is-the-budget-game-played-with-ai-tokens-24fa872c86bf>
- [10] <https://blog.exceeds.ai/enterprise-ai-coding-roi-studies/>
- [11] <https://www.index.dev/blog/ai-tool-roi-statistics-for-engineering-teams>
- [12a] <https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/#methodology>
- [12b] <https://metr.org/blog/2026-02-24-uplift-update/#wider-adoption-of-ai-has-made-it-more-difficult-to-measure-task-level-productivity>
- [13] <https://www.humai.blog/uber-burned-its-entire-2026-ai-budget-in-four-months-claude-code-did-it/>

▼ Brainstorming

Motivating Context: Uber recently burnt through its entire AI budget for 2026 in April, just on token spend alone, after the company purchased Claude Code licenses for its roughly 5000 engineers in December. To an extent, this was of their own making, after the company's heads released an 'internal leaderboard' to track and rank engineers based on their amount of Claude Code usage [1].

This tracks a recent, amusing, internal trend in Silicon Valley where other companies like Meta and Shopify, eager to accelerate productivity per employee, have begun encouraging teams and engineers to outcompete each other on 'AI use'/token spend, viewing token consumption as a proxy for the amount of internal AI adoption, itself a proxy for productivity [8].

▼ Outline

- Uber as canary in the coal mine: spent entire 2026 AI budget within first 4 months of the year
- **Context:** executives and company heads, eager not to be left behind in the AI race, are racing to encourage internal AI adoption and use in their companies
- **Setup 1:** As part of this, many companies have purchased massive enterprise licenses for coding tools/agents (like Claude Code, Codex, etc.) for their employees.
- **Setup 2 (Goodharting):** companies are starting to treat an employee's 'amount of AI spend' as like a rough proxy for productivity.
 - Since AI helps one get work done, it reasons that the more one uses AI, the more work is getting done.
- **Steelman (Productivity):** the premise is not itself nonsensical. By one estimate from Index, roles like software developers save, on average, between 40-65% of the full time they would have taken on tasks [11]. Another claims that developers who use AI assistants for coding save "an average of 2 to 3 hours per week", with some power users saving more than 5 hours per week [10].
 - **Qualification:** But these claims are usually made from outside analyses of internal self-reported data from companies. What happens when we look at direct evidence instead?
 - A July 2025 report from METR showed a surprising result: contrary to expert forecasts and software developer's own estimates, coding with AI agents appeared to slow productivity and cause developers to take about 19% longer on programming tasks [12a]
 - **Meta-Qualification:** But the evidence here seems ambiguous, possibly because the technology is also fast-moving; by 2026, METR had soft-redacted their earlier claim out of worries about selection bias and measurement and study design errors. Their new study now estimates an average 'speed-up' effect of 18% from developers who use coding agents [12b].
- ▼ **Problem (As Identified in Current Debates/Lit Review):** It is unclear whether any of this spending on AI, along with all of its necessary implications and externalities (datacenter buildout, more demand for energy, water consumption in open cycle cooling systems), will actually translate into meaningful productivity gains for the companies that use them.
 - *"The industry's standard response to consumption-cost stories is that artificial intelligence pays for itself in productivity gains. Uber's case complicates that argument. The marginal productivity gain from a senior engineer running agentic workflows has to clear a much higher token-cost hurdle than the gain from an engineer running autocompletes. Five-to-twenty-fold increases in per-developer consumption are now documented in agentic mode, and no public benchmark shows a matching multiplier on output value."* - Forbes [1]
 - *"The productivity numbers were real: approximately 11% of all pull requests at Uber are now opened by AI agents, and around 11% of live backend code updates are written by AI systems handling everything from ride-matching algorithms to dynamic pricing and bug fixes. AI-related costs, however, rose roughly 6x since 2024, and the full-year budget Naga had set aside for AI tooling was gone before the second quarter started."* - Humai [13]
 - *"Productivity gains vary widely across teams because they depend on task type, experience level, and review practices. Measured data shows steady improvements, but not the extreme gains often*

claimed in marketing. [...] Real organizations report only 0.3 to 1x productivity improvement, which is far lower than the common 2x productivity claims made by AI tool vendors." - Index [11]

- *"Spending more doesn't mean producing more. Most companies handing out AI access to employees have no real idea what it's being used for, whether it changed any outcomes, or whether those outcomes would have been the same anyway."* - Shruti Gandhi [3]
- *"Uber COO Andrew Macdonald set off a discussion about a potential reckoning over AI costs when he said the tech giant wasn't seeing productivity gains relative to its overall bill. "That link is not there yet, right?" he told Rapid Response. "I think maybe implicitly there is more that is getting shipped, but it's very hard to draw a line between one of those stats and, 'Okay, now we're actually producing 25% more useful consumer features.'""* - Business Insider [3]
- *"AI commentator Gary Marcus said that "agent hysteria" drove a lot of initial AI demand. "Now the critical variable is whether these agents, which are expensive, pay off," Marcus wrote on X in May. "The cases of Uber and Microsoft, the frustrations of GeoHotz (who was all in), etc are possibly signs that the reliability and payoff may not be there for a lot of customers." "If enough other companies report the same, the bubble pops. 🌪️," he wrote on X."* - Business Insider [3]
- *"In January, Accenture told senior staff that they must regularly use AI tools to be considered for promotions. As an anonymous Accenture employee told Jacobin, "I wish there was more transparency on the enforcement of AI usage across senior staff. The policy is vague. I am worried it'll encourage people to just use AI for the sake of hitting some target." [...]"*

The use value of AI and its costs to capital have diverged. The productivity gains are likely real but insufficient. For example, per-developer consumption at Uber has increased five- to twentyfold, but no public benchmarks show a matching increase in "output value." According to the LLM researcher, "whether you're a model provider or tech company in the AI economy, the competition is fierce and you feel pressure to keep the spending up, even if you're investing more than you should or can."" - Sophie Bandarkar, Jacobin [4]

- *"Tokenmaxxing is the same game with a different currency. Meta and Shopify now include AI usage in performance reviews. Jensen Huang said he'd be "deeply alarmed" if an engineer with a \$500K salary didn't consume at least \$250K in tokens within a year. Meta's CTO has publicly encouraged engineers to spend the equivalent of their salaries on tokens, claiming 10x productivity gains."*

The message to the rank and file is clear: if you're not burning tokens, you're falling behind. And just like the budget game, people respond to the incentive, not the intent. Running five parallel agents overnight produces nothing that ships, but it produces a ranking. The leaderboard rewards what it can see." - Willian Corrêa [9]

- *"Jen Stave, executive director of the Harvard Business School AI Institute, hears the same from enterprise leaders. "I've talked to a dozen CTOs or CIOs who are all saying, 'Actually I'm having a really hard time finding an ROI framework for this,'" she said."* - CNBC [8]
- *"And yet, profits remain scarce, and some economists are warning of an impending bubble. That won't stop leader OpenAI from striking multibillion-dollar chip deals with Nvidia, AMD, and most recently, Broadcom. Harvard Business School's Andy Wu says that while AI's promise is clear, the potential return on these massive investments remains murky."* - Danielle Kost, HBS [5]

- *"Moreover, especially in agentic coding, there are massive losses, up and down the food chain. The providers (Open AI, Anthropic, Cursor, etc) lose tons of money, and are scrambling to raise prices, Customers are balking at the new usage models. (In a new analysis, Gerben Wierda has argued that Anthropic and OpenAI "may actually pay \$1000 for every \$100 you pay them". [...]) It could well be that once the providers (Open AI, Anthropic, Cursor, etc) attempt to charge enough money to cover their losses, the cost of AI could become more expensive than the humans it is replacing."* - Gary Marcus, Substack [6]

Problem (Original Contribution)

Regardless of whether companies are able to construct strong-enough governance frameworks to ensure AI spend corresponds to meaningful use, there is a larger question lurking underneath.

- (1) If current levels of CapEx and spending from **frontier AI labs** are dependent on continued, predicted growth in demand for their models and AI services from **other companies**;
- (2) And this increase in **enterprise demand** is premised on the expectation that AI (or AGI) will allow these **other companies**:
 - (2a) to develop, and offer, better goods and services,
 - (2b) possibly even at lower prices due to automation reducing labor costs;
- (3) thereby generating more **consumer demand** for these **other companies'** goods and services, who then;
- (4) see the direct gains and return on investments into AI-related services from the improvements to their company/products/services;
- (5) causing these **other companies** to continue increasing their **enterprise demand** for, and their use of, AI products, in order to **further improve their goods and services** so that they can keep increasing **consumer demand**;
- (6) thereby giving **frontier labs** the funding source needed to continue research and development into progress towards AGI and eventually recouping initial spending on pretraining and datacenter infrastructure.

Problem Statement

What if the economic gains for firms from AI *are real* (4), but the ability for firms/other companies to continue improving their goods and services to increase consumer demand (5) *isn't*, because there is some ceiling to how much any product/service can be improved, and once most companies hit that point, it becomes irrational to continue spending on AI—at least in that moment?

Intuition Pump

- Take the case of Uber, which exhausted its entirely 2026 AI budget in the first 4 months. At an abstract, philosophical 'high-level', what does it *really* even mean for Uber to become 'better'?
- Uber is a ride-hailing app/company. They connect drivers (contractors) to users (customers). The main advantage they had over traditional taxi companies was providing real-time and geographic information about where their drivers were with respect to any given customer.

- Now imagine Uber's SWEs spend \$500,000 on AI tokens to increase their output and maximize productivity gains. Ask yourself:
 - What is this 'output'?
 - What are these 'gains' and what does it mean/look like for them to be maximized?
- In the most generous interpretation, let us say the state of the art ("SOTA") models in late 2026 are so capable and intelligent that they come up with radically new powerful computational methods for solving optimization problems like the Travelling Salesman problem, and help Uber invent a new path-optimization algorithm that makes everything 10x more efficient on the backend.
- Great. Now what?
- If they continue spending on AI, hypothetically, what gains could they expect to receive?

1. How much better could Uber (or the app) really become?

It's an app. You tap buttons. Genuinely, in the most creative way possible, what gains are even possible?

- Loading Times: maybe they optimize the app to take up less RAM and start faster?
 - But how long does the app already take on your phone to start-up?
 - 4 seconds? 6 seconds?
 - If they can 'cut it down' to 2 seconds, how much does that improve the consumer experience and how much are consumers willing to pay for that? What are the actual gains?
 - But once they've optimized the app, what next? There aren't an infinite number of suboptimal design decisions and programming failures to improve upon—or rather, if there are, their initial, actual human SWEs did a terrible job and are likely going to all be fired now.
 - **My point**: the gains in this area are real, but finite. You can't keep making any arbitrary app/program faster. Computational theory establishes lower bounds on how fast any processes involving a fetch-read-write cycle can become.
- UI and Design: well maybe they could make the app prettier nicer, not just in some minor aesthetic change sort of way but a serious design overhaul centered around the latest empirical research and findings from human-computer interaction?
 - Maybe this new, prettier version of the app is not just nicer to look at, but so much more pleasant to use that it genuinely somehow increases weekly usage for the same fixed pool of users by 1-5%.
 - Perhaps they add a fun, addictive mini-game to the app when waiting for a ride which turns out to be as fun as actual AAA games.
 - Maybe they change the color scheme and the entire's app design to create a kind of feeling of 'artificial pressure' so that when users are on the app, they are likelier to just accept whatever price shows up first for a ride—rather than waiting, and re-opening/checking the app several times to try and see if prices fall.
 - **Problem**: again, if this works, great. But then what? Will there be an 'even more addictive color scheme' next? And will there always be one? Or are the space of improvements here

finite.

- Automation and Driverless Cars: perhaps AI eventually invents reliable driverless cars that are cheaper to manufacture AND cheaper to operate and service over a 10-year span than using human drivers.
 - This is great, except, Uber already doesn't care about 'cheaper to service' because remember, drivers (in most jurisdictions) are contractors, so they pay for their own vehicles and any repairs or servicing that is needed.
 - And Uber also doesn't own any of the cars that their drivers use, so these driverless cars being '*cheaper to manufacture*' alone does nothing because the real competition is between:
 - purchasing a driverless car *vs.* using a car that someone else already owns (free)
 - But let's say Uber does automate their entire fleet and replace all human drivers with self-driving cars.
 - **Problem:** You cannot do this recursively. Once the humans have been replaced with automated vehicles, they have been replaced. Again then, any amount of AI-related spending in this area seems more likely than not to be finite (within some meaningful interval of time).
 - You might argue, but won't the self-driving cars themselves improve over time giving Uber reason to continue spending on automated vehicles?
 - But this depends on the actual gains. If the second generation of automated vehicles shows a 68% reduction in traffic accidents per 1M miles, but costs 25% more than the first generation, and the number of accidents in absolute terms is already some small figure (say, 9 accidents per year) such that everyday consumers do not see it as a factor in the company's reputation, why would they bother spending more for such little gains?

▼ The Extent of Finite Improvements in The Economy

Now what if cases like Uber are not the exceptions but the rule.

It seems plausible, *prima facie*, that for most firms in the economy, whatever their niche/market/sector, most goods and services face natural 'ceilings' on improvement, either because:

1. The number/space of possible improvements at a given level of technology has a ceiling.
2. The rate of improvement for any technology, *at any given level of technology*, has no ceiling, **but**,
 - a. there are decreasing marginal returns, which at some point eclipse any justification for continued, increasing spending on investment or use of AI products and services

Another Intuition Pump

For sake of argument, let us explore theoretically some businesses and their use cases for AI to hypothesize about the likelihood of infinite, or at least, ***continually recursive*** within some meaningful time interval (say 5-15 years)

- **Book Stores**
 - It's a book store. They sell books. Maybe they can come up with better marketing strategies to make more of the population want to buy books, or make their existing pool of customers want

to buy more books per customer?

- But how possible is it for this to be recursive?
- Consider also Red Queen dynamics: if one book store invents a radically more effective marketing technique, it is unlikely to act as a 'moat' for long. If all book stores start running better ads, but the cost of getting a book out to print does not fall per unit (because they are sold for the most marginal level of profit), for the same amount of fixed spending power, all that happens is:
 - **(substitution effect)**: for some time, existing customers buy more books from any given book store, by diverting discretionary spending on other things.
 - **(medium-run)**: book stores spend more per quarter on marketing (via AI) than they did before to compete with each other for purchases, **but**, the rate of growth in demand for books sees only a level change
 - *i.e.* each customer buys +6 more books each year than they did previously, but this stays fixed—as opposed to customers buying +6 more books every year than the year they did before.
- **Retail Clothing Stores**
 - Main cost factor here is production inputs and per unit productivity for items
 - Main revenue factor here is (1) cultural tastes/social dynamics, and (2) consumers' discretionary spending power
 - Re: (1) cultural tastes
 - Again, through improved marketing strategies that make better use of latest SOTA insights from psychology and behavioural economics, it may be possible that clothing stores can run better large scale ad campaigns to induce greater demand across society for their products or even manufacture demand for new products/clothes via new trends.
 - Theoretically, this specific mechanism may actually be infinitely recursive: it is possible that there is always a 'more optimal' or 'more convincing/addictive' marketing strategy that is, *ceteris paribus*, more likely to cause a given ad view to convert into a sale.
 - But the real constraint here is like a mixture of (2) + Red Queen Dynamics
 - If newer AI models can come up with better ads for companies, **but** all companies have access to these models, and so all companies can arbitrarily improve their ad strategies up to the performance of the current SOTA in marketing strategies,
 - **AND the spending power of the average consumer remains fixed**—say, because in the most optimistic case, while automation does not 'destroy' jobs, it prevents new ones from being created or prevents most people from ascending up the career ladder since human labour is only competitive for low per unit/hour cost tasks
 - **AND** the nature of physical goods is that they take up space to store, even when unused, meaning most potential customers have a physically-mediated upper bound on amount of consumption, regardless of true preferences/demand
 - Then again, 'improvements' from AI seem unlikely to be recursive. Although in this case, continual spending on AI is possible (because of race dynamics), but the dominant result is mostly to shift capital from those firms to AI companies, rather than resulting in gains to the

clothing stores themselves, because a firm must spend more to be at the same level of 'marketing power' as competitors, but the:

1. **Pool/number of customers (population level) remains fixed or does not increase**
2. **Spending power of consumers remains fixed**
3. **Amount of possible consumption per potential customer is bounded by the size of their living space/closet**

And so on, and so forth. The extent of this 'no need or even possibility for infinite improvements' mechanism is left as an exercise for the reader.

Research Question

1. How likely are the gains from automation—for most firm—to exhibit recurrent, infinitely-extendable properties?
2. What kinds of firms operate in sectors where products and services *can always be improved*?
3. What proportion of the economy do the firms in (2) make up?
4. Even if AI technology itself can be recursively improved, is this true of other existing technologies, and are the marginal returns from improvements to these other technologies constant, or if increasing, increase slowly enough to continue justify current levels of AI spending and use?
5. If (4) is not true, does this also derail continued progress on AI via a lack of growth and spending on AI companies products and services?

▼ Possible Methodology

- analysis of time series data
 - use to determine duration and magnitude of impacts on firm-level growth and output from past 'new technologies'
- lagged effects model + impulse responses — as per, Ostry et Berg (2017)
- consider non-parametric models?
- or curve/data-fitting using a framework with minimal inductive biases? like FPs — as per, Dash, Ferris, and Voia (2023)
- treat as a constrained optimization problem by representing the average level of change in historical data as a prediction of maximized expectation, allowing use of Lagrange Multipliers under Karush-Kuhn-Tucker conditions (to constrain problem space) — per, Andreani, Martinez, and Schuverdt (2005)

▼ Proposed Plan of Action

1. Find and make a shortlist of long-run, time-series candidate datasets that contain information applicable for this issue.
2. Study candidate datasets to determine:
 - a. limitations in data, and concomitant limitations on types of analysis possible
 - b. feasibility of cross-integrating, referencing data from the different datasets
 - (*i.e.* do some provide quarterly data while others only annual? what methods can be used to salvage this? reweighting perhaps?)
 - do the datasets use the same units of [anything], *i.e.* what currencies are financial figures in, are they real or nominal, and if real, what is the base year and what method was used? (*e.g.* chain-fisher?)
3. Determine if the structure of the problem and data availability permits causal inference techniques.
 - a. Do we have enough data, and the right kind of data—*e.g.* synthetic controls or synthetic random assignment—via regression discontinuity, instrumental variables?
4. **If not**, don't despair. Accept and simply treat as a correlation inference/pure regression problem. Now determine:
 - a. What kind of regression methods to use?
 - b. What are acceptable upper and lower bounds on SEs and Type I, II, errors?
 - c. What robustness tests are possible?
5. Clean and format the data to be ready for analysis.
6. Formalize a model for the null and alternative hypothesis, w.r.t. the datasets available and the kinds of methods chosen in either (3) or (4).
7. (Finally) Test model against data and don't be a bitch and make sure you report all findings even if null and make you look like a complete idiot.
8. Draft LW post??

▼ Actual Draft (Writing)

Context

In December 2025, Uber rolled out Claude Code to software engineers across its organization to boost productivity. "Adoption [of agentic coding tools] climbed from 32% of engineers in February to 84% by March." By Spring of (early) 2026, this had reached 95%, and "roughly 70% of committed code" originated from AI agents.

According to Forbes, the monthly amount spent on AI per engineer "ranged from \$150 to \$250 on average", with power users closer to \$500–\$2000.